Abhinav Ravindra Dehadrai

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Education

PhD, *August 2018* (thesis submitted), CPI: 8.29,
Mechanics & Applied Mathematics group,
Mechanical Engineering, Indian Institute of Technology, Kanpur, India.
Title: *Dynamics of Cables in Two Examples: Towing of Underslung Loads and Vertically Traveling Beams.*
Advisers: Ishan Sharma, Shakti Singh Gupta

MTech, *August 2010*, CPI: 6.75
Mechanical Engineering,
Indian Institute of Technology, Kanpur, India.
Thesis: *Modeling of liquid fuel sloshing in fuel tanks of spacecrafts.*
Adviser: Ishan Sharma.

BE, *June 2006*, Avg: 56.93%
Mechanical Engineering,
Yeshwantrao Chavan College of Engineering, Nagpur University, India.

Research

PhD

• Dynamics of cable towing a ring-shaped underslung load.

The towing of a large, ring-shaped, underslung load attached by a heavy, flexible cable is investigated. For surveying large areas in less time, a large ring-shaped water/mineral detector is suspended from a helicopter with the help of long cable. For optimal operation the ring-shaped detector should be in the horizontal plane, i.e. normal to gravity. The helicopter's motion and aerodynamic forces affect the motion of the cable and the detector. Further, these forces may misalign the detector or even destabilize it during operating conditions. We model the cable as geometrically exact (GE) beam incorporating large displacements and rotations, while the detector is modeled as rigid body. The equations of motion are nonlinear in displacements and rotation parameters. We developed a nonlinear finite element (FE) model and computed the response of the cable-detector system to aerodynamic forces. The dynamics of system is investigated during hovering, forward and maneuvered flight conditions.

• Dynamics of vertically traveling cable.

Traveling cables are fundamental driving mechanisms in elevators, conveyor belts, automobile chain-drives, cableways, etc. In these, all or part of the cable is inclined to gravity. During operations the cable tends to oscillate transversely as it travels longitudinally. In this study, the cable is modeled as a linear elastic beam that resists bending moment and shear, and as an inextensible string which supports only tension. The dynamics of horizontally traveling (traveling direction normal to gravity) beams and strings is well studied. We investigate the influence of end-tension, bending stiffness, slenderness ratio, external damping, and gravity on the stability of cables which travel vertically. We found that a vertically traveling cable becomes unstable at lesser speed than a horizontally traveling cable. This work is recently published in the ASME Journal of Computational & Nonlinear Dynamics.

MTech

Modeling of liquid fuel sloshing in a spacecraft's fuel tank.

Sloshing refers to the motion of a liquid inside a partially-filled container. Liquid propellants constitute the largest part of the total mass of many spacecrafts. Sloshing of liquid fuel in the fuel tanks affects the orientation and angular velocity of the spacecraft. The deviation of the

actual attitudinal motion from the predicted rigid body motion, may allow us to know the life remaining in the spacecraft provided we are able to quantify, to some extent, the effects of fuel sloshing on the spacecraft's motion. We approximate the motion of the fuel in its first mode by the motion of a spherical pendulum whose length and mass are related to the amount of fuel remaining. The spacecraft is modeled as a rigid body.

Journal Publications

• Dehadrai A.R., Sharma I., Gupta S.S., Stability of Vertically Traveling, Pre-tensioned, Heavy Cables. *ASME. J. Comput. Nonlinear Dynam.* 2018; 13(8):081003-081003-9. doi:10.1115/1.4040344.

• Dehadrai A.R., Sharma I., Gupta S.S., Dynamics of cable towing a large ring-shaped slung load. (in preparation)

Seminar Talks

I Dynamics of under-slung load system & Fluid-structure interaction (FSI)
• APM-2015, St. Petersburg (Russia)
• Helix-2015 and FSI workshop, Hyeres (France)
• ESMC-2015, Madrid (Spain)

II Dynamics of traveling tensioned heavy cable
• ICCMS-2012, Hyderabad (India)
• APM-2014, St. Petersburg (Russia)
• ISTAM-2014, Bangalore (India)

Relevant Courses

• Dynamics and vibrations • Continuum mechanics • Vibrations of continuous systems • Applied mathematics • Asymptotic and Perturbation methods • Spectral methods for partial differential equations • Numerical solutions to integral equations • Nonlinear dynamics. • Theory of elasticity

Academic Experience

Teaching assistant/ Tutoring:
• Dynamics and Vibration • Mechanics of solids • Continuum mechanics • Theory of machines and mechanisms.

Coding Environments

Matlab, Maple, and Abaqus (Basic FEM).